

Jack Cashman

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EDUCATION

•Purdue University

Ph.D. in Electrical and Computer Engineering

Starting August 2026

West Lafayette, IN

•University of Rochester

B.S. in Computer Science (Honors Research)

August 2022 – May 2026

Rochester, NY

RESEARCH & TEACHING EXPERIENCE

•University of Utah

Research Assistant under Prof. Ponnuswamy Sadayappan

May 2025 – Present

Salt Lake City, UT

- Investigated sparse matrix reordering schema cache hierarchy utilization and its effects on Sparse Matrix Vector Multiplication (SpMV) performance.
- Reduced error of a machine learning model for pairwise ranking of data layouts based on predicted L1D, L2, and L3 cache misses, addressing the fragility of traditional reordering heuristics.
- Conducted controlled experiments by disabling hardware prefetching and varying compiler optimization levels to isolate performance effects and align empirical results with model predictions.

•University of Rochester, Dept. of Computer Science

Honors Thesis Researcher

August 2025 – Present

Rochester, NY

- Designing an analytical cache model that directly targets performance factors including cache capacity, associativity, latency, replacement policy, prefetchers, and thread count.
- Integrating analytical prediction with genetic search and heuristics to drive sparse matrix reordering decisions, departing from traditional graph theory approaches.
- Parallelizing reordering search and analytical modeling for improved speed of reordering, and validating results with hardware counters.

•University of Rochester

Research Assistant under Prof. Chen Ding

May 2024 – Present

Rochester, NY

- **Loop Asymptotic Locality Analysis (LALA):** Developed a symbolic tool to model the reuse distance distribution of affine programs, reducing analysis time from doubly exponential to linear relative to program size.
- **Zipfian Caching for Web Workloads:** Designed and evaluated caching policies to outperform LRU on dynamic Zipfian web workloads, achieving a faster convergence to the optimal hit rate.
- **LLM Text Detection:** Used locality measurements of data movement distance for varying N-gram text representation to classify text as human vs. LLM-generated.

•University of Rochester

Teaching Assistant, Computer Organization

August 2024 – Present

Rochester, NY

- Led study sessions and office hours to help undergraduate students understand core systems concepts, including memory hierarchy, instruction sets, and digital logic.
- Wrote, graded, and proctored exams, and provided feedback on assembly language and C programming assignments.

SELECTED PROJECTS

•Tiny-QMoE LLM Compression System

Course Research Project, Machine Learning Systems

Fall 2024

- Engineered a quantization and dictionary-based compression pipeline to enable inference of large language models (LLMs) on memory-constrained devices (e.g., mobile phones).
- Achieved **23–35x compression** on LLaMA models with minimal accuracy loss on MMLU and ARC benchmarks.
- Gained hands-on experience with GPU memory hierarchy, low-latency compression algorithms, and LLM benchmarking.

•Latency Based Tiling

Spring 2025

Course Research Project, Honors Undergraduate Problem Seminar

- Built a triangle loop based cache latency micro benchmark for approximating cache capacities for L1D, L2, and L3 caches.
- Leveraged the cache approximation for polyhedral program tiling to produce a 21% performance speedup with effectively no compilation overhead unlike auto-tuning.

•Mid-End Compiler Optimization Framework

Spring 2025

Course Project, Advanced Compilers (Software Analysis & Improvement)

- Built a compiler optimization framework from scratch, operating on an intermediate representation.
- Implemented classic and advanced optimizations including dead code elimination, partial redundancy elimination, and constant propagation using abstract interpretation.

PUBLICATIONS

- **Preprint:** Jack Cashman. *Latency Based Tiling*. arXiv preprint arXiv:2510.15912, 2025.
- **Preprint:** Jack Cashman, Jiaqi Nie. *Tiny-QMoE*. arXiv preprint arXiv:2509.22951, 2025.
- **In Preparation:** Sparse matrix reordering research with Prof. Ponnuswamy Sadayappan.
- **In Preparation:** LALA (Loop Asymptotic Locality Analysis) with Prof. Chen Ding.

TECHNICAL SKILLS AND INTERESTS

Languages: Rust, C, RISC-V Assembly, x86-64 Assembly, Coq, CUDA

Developer Tools: Git, Linux/Unix Shell, VTune, PAPI, LIKWID, Perf, GDB

Relevant Coursework: Advanced Compiler Optimizations, Machine Learning Systems, Computer Organization, Operating Systems, Programming Language Design, Data Structures & Algorithms, Computer Models & Limitations, Honors Research Problem Seminar, Computer Networks

Areas of Interest: Compiler Optimization, High Performance Computing, Locality Theory, Cache Modeling, Computer Architecture, Type Systems/Safety, Machine Verified Proofs, Side Channel Attacks